

Assignment 1 Solutions

Problem 1: Adjacency Matrix

Question: Consider the following adjacency matrix of a network.

$$\begin{pmatrix} 0 & 0 & 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}$$

- (a) Is the network directed or undirected? (Explain why).
- (b) Draw the network.
- (c) What are the in-degree and out-degree of each node?
- (d) Determine the in-degree distribution and the out-degree distribution.

Solution:

- (a) The network is **directed** because the adjacency matrix is asymmetric ($A_{ij} \neq A_{ji}$ for some i, j).
- (b) [Network diagram not shown here]
- (c) Node degrees:

Node 1: in-degree = 1, out-degree = 3

Node 2: in-degree = 1, out-degree = 2

Node 3: in-degree = 2, out-degree = 2

Node 4: in-degree = 3, out-degree = 1

Node 5: in-degree = 1, out-degree = 1

Node 6: in-degree = 2, out-degree = 1 (d) In-degree distribution:
 $P(0) = 0, P(1) = \frac{1}{2}, P(2) = \frac{1}{3}, P(3) = \frac{1}{6}, P(k) = 0$ for $k > 3$
Out-degree distribution:
Same as in-degree distribution.

Key Concepts:

Adjacency matrix: $A_{ij} = 1$ if edge from i to j

In-degree: $k_i^{\text{in}} = \sum_j A_{ji}$ ()

Out-degree: $k_i^{\text{out}} = \sum_j A_{ij}$ ()

Degree distribution: $P(k)$ = fraction of nodes with degree k

Problem 2: Diameter, Eccentricity and Radius

Question: Calculate the diameter D of the following networks (size N):

- (a) Fully connected network
- (b) Star network
- (c) Linear chain
- (d) Square lattice with L nodes per side
- (e) Show (d) approaches $2\sqrt{N}$ as $N \gg 1$
- (f) d -dimensional hypercubic lattice
- (g) Cayley tree with branching factor k
- (h) Find eccentricity and radius of linear chain

Solution:

- (a) $D = 1$ (all nodes directly connected)
- (b) $D = 2$ (max distance through center)
- (c) $D = N - 1$
- (d) $D = 2(L - 1)$
- (e) $N = L^2$, so $D = 2(\sqrt{N} - 1) \approx 2\sqrt{N}$ for $N \gg 1$
- (f) $D \approx d\sqrt{N}$
- (g) Nodes reachable in d steps: $k(k - 1)^{d-1}$

Total nodes: $N = 1 + k \frac{(k-1)^d - 1}{k-2}$

Diameter: $D = 2d = 2 \frac{\ln[N(k-2)+2] - \ln k}{\ln(k-1)}$

(h) Eccentricity of node i : $\max(i, N - 1 - i)$

Radius: $\frac{N-1}{2}$ if N odd, $\frac{N}{2}$ if N even

Key Concepts:

Diameter: $D = \max_{i,j} d(i, j)$ ()

Eccentricity: $e(v) = \max_u d(v, u)$ ()

Radius: $r(G) = \min_v e(v)$ ()

Distance in lattices: scales with dimension

Cayley tree: $N \propto (k - 1)^d$ (exponential growth)

Problem 3: Bipartite Matrix

Question:

- (a) For bipartite network with n_1, n_2 nodes:
 - (i) Max number of links?
 - (ii) How many links of complete network cannot occur?
 - (iii) Show $c_2 = \frac{n_1}{n_2} c_1$

(b) Find incidence matrix and one-mode projections

Solution:

(a)(i) Max links = $n_1 \times n_2$

(ii) Complete network links = $\frac{1}{2}(n_1 + n_2)(n_1 + n_2 - 1)$

Missing links = $\frac{1}{2}[n_1(n_1 - 1) + n_2(n_2 - 1)]$

(iii) Since total links equal: $n_1 c_1 = n_2 c_2 \Rightarrow c_2 = \frac{n_1}{n_2} c_1$

(b) Incidence matrix:

$$B = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

[One-mode projections not shown]

Key Concepts:

Bipartite network: nodes in two disjoint sets, edges only between sets ()

Incidence matrix: $B_{i\alpha} = 1$ if node i connects to node α in other set

One-mode projection: connects nodes that share neighbors in original bipartite network

Mean degree relation: $n_1 c_1 = n_2 c_2$

Problem 4: Spectrum of Star Graphs

Question: For a star graph with n nodes:

- (a) Average degree?
- (b) Eigenvalues of adjacency matrix
- (c) Eigenvalues of Laplacian matrix

Solution:

(a) Average degree = $\frac{2(n-1)}{n} = 2 - \frac{2}{n}$

(b) Adjacency eigenvalues:

$\lambda_1 = \sqrt{n-1}, \lambda_2 = \dots = \lambda_{n-1} = 0, \lambda_n = -\sqrt{n-1}$

(c) Laplacian eigenvalues:

$\lambda_1 = n, \lambda_2 = \dots = \lambda_{n-1} = 1, \lambda_n = 0$

$\lambda_2 = 1 > 0 \Rightarrow$ graph is connected

Key Concepts:

Average degree: $\langle k \rangle = \frac{2L}{N}$

Adjacency matrix: $A_{ij} = 1$ if edge between i and j

Laplacian matrix: $L = D - A$ where D is degree matrix

Algebraic connectivity: $\lambda_2(L) > 0$ iff graph is connected

Eigenvalues: solve $\det(A - \lambda I) = 0$

Assignment 2 Solutions

Problem 1: Matrix Formalism

Question: Show that for adjacency matrix A :

- (a) Degree vector $\mathbf{k} = A\mathbf{1}$
- (b) Number of links $L = \frac{1}{2}\mathbf{1}^T A\mathbf{1}$
- (c) Common neighbors matrix $C = A^2$

Solution:

(a) $k_i = \sum_j A_{ij} = \sum_j A_{ij} 1_j \Rightarrow \mathbf{k} = A\mathbf{1}$

(b) $L = \frac{1}{2} \sum_i \sum_j A_{ij} = \frac{1}{2} \sum_i \sum_j 1_i A_{ij} 1_j \Rightarrow L = \frac{1}{2} \mathbf{1}^T A\mathbf{1}$

(c) $C_{ij} = \sum_r A_{ir} A_{rj} \Rightarrow C = A^2$

Key Concepts:

Degree vector: $\mathbf{k} = A\mathbf{1}$ ()

Link counting: $L = \frac{1}{2} \sum_i k_i = \frac{1}{2} \mathbf{1}^T A\mathbf{1}$

Common neighbors: $C_{ij} = \sum_r A_{ir} A_{rj}$ (number of paths of length 2)

Matrix multiplication: $(A^2)_{ij} = \sum_k A_{ik} A_{kj}$

Problem 2: Centrality in Directed Network

Question: For network with adjacency matrix

$$A = \begin{pmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}$$

- (a) Find in/out-degree sequences
- (b) Find in/out/undirected degree distributions
- (c) Calculate Katz centrality $\mathbf{x} = \beta(I - \alpha A)^{-1} \mathbf{1}$

Solution:

(a) In-degree: $\{0, 2, 1, 1\}$

Out-degree: $\{2, 0, 1, 1\}$

(b) In-degree distribution: $P_{\text{in}}(0) = \frac{1}{4}$, $P_{\text{in}}(1) = \frac{1}{2}$, $P_{\text{in}}(2) = \frac{1}{4}$

Out-degree distribution: same as in-degree

Undirected degree: $P(2) = 1$

(c) $A^0 = I$, $A^1 = A$, $A^2 = \begin{pmatrix} 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$, $A^n = 0$ for $n \geq 3$

Katz centrality:

$$\mathbf{x} = \beta \begin{pmatrix} 1 + 2\alpha^2 \\ 1 + \alpha \\ 1 \\ 1 + \alpha \end{pmatrix}$$

Key Concepts:

Katz centrality: $\mathbf{x} = \beta(I - \alpha A)^{-1} \mathbf{1} = \beta \sum_{n=0}^{\infty} \alpha^n A^n \mathbf{1}$ ()

Matrix series: converges if $\alpha < 1/\lambda_{\max}(A)$

Directed degrees: in-degree vs out-degree ()

Problem 3: Closeness Centrality

Question: For an undirected tree with edge between nodes 1 and 2 dividing it into regions of n_1 and n_2 nodes, show:

$$\frac{1}{c_1} + \frac{n_1}{n} = \frac{1}{c_2} + \frac{n_2}{n}$$

Solution:

For any node j :

$$d_{1j} - d_{2j} = \begin{cases} -1 & j \in R_1 \\ 1 & j \in R_2 \end{cases}$$

Using definition $\frac{n}{c_i} = \sum_j d_{ij}$:

$$\frac{n}{c_1} - \frac{n}{c_2} = \sum_j (d_{1j} - d_{2j}) = n_2 - n_1$$

Rearranging gives the required relation.

Key Concepts:

Closeness centrality: $c_i = \frac{n-1}{\sum_j d_{ij}}$ ()

Tree properties: unique path between any nodes

Distance relation: $d_{1j} = d_{2j} \pm 1$ depending on region

Problem 4: Betweenness Centrality

Question:

(a) For a vertex with degree k dividing tree into regions of size $n_1, \dots, n_k$, show unnormalized betweenness:

$$x = \frac{1}{2} \left[n^2 - \sum_{m=1}^k n_m^2 \right]$$

(b) Find betweenness of i th vertex in line graph

Solution:

(a) Betweenness counts paths through vertex:

$$x = \sum_{i \neq j} n_i n_j = \frac{1}{2} \left[\left(\sum_i n_i \right)^2 - \sum_i n_i^2 \right] = \frac{1}{2} \left[n^2 - \sum_{m=1}^k n_m^2 \right]$$

(b) For i th vertex in line of n nodes:

Left region: $i - 1$ nodes, right region: $n - i$ nodes

Betweenness: $x = (i - 1)(n - i)$

Key Concepts:

Betweenness centrality: fraction of shortest paths through node ()

Unnormalized betweenness: counts total paths through node

Tree property: unique shortest path between nodes

Region sizes: $n = \sum_i n_i$

Problem 5: Strong and Weak Ties

Question:

(a) How may edge between B and C be labeled if all nodes satisfy Strong Triadic Closure?

(b) Which nodes violate Strong Triadic Closure?

Solution:

(a) Edge BC must be weak tie (W) or absent. If strong (S), would require edges between C-E and B-F.

(b) Node C violates: has strong ties to B and E but no edge B-E.

Node E violates: has strong ties to C and D but no edge C-D.

Key Concepts:

Strong Triadic Closure: if A has strong ties to B and C, then B and C must have at least a weak tie ()

Strong tie: close relationship (family, close friends)

Weak tie: distant relationship (acquaintances)

Bridge: edge whose removal increases components (weak ties often bridges)